

Predicting human mobility flows in cities using deep learning on satellite imagery

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Understanding the interaction between complex urban environments and human mobility flow patterns underpins adaptive transport systems, resilient communities, and sustainable urban developments, yet inter-regional origin-destination mobility flow information from traditional surveys are costly to update. The satellite imagery offers up-to-date information on urban sensing and opens avenues to examine urban morphology-mobility dynamics. This study develops a deep learning model, Imagery2Flow for predicting fine-grained human mobility flows in urban areas using 10 to 30-meter medium resolution satellite imagery in a timely and low-cost manner. Extensive experiments demonstrate good performance and flexible spatial-temporal generalizability on the top-10 largest metropolitan statistical areas of the United States. Through exploring the spatial heterogeneous effects, we investigate the urban factors (centrality and compactness) influencing human movement flow distributions, enhancing our comprehension of their interactions. The spatial transferability of Imagery2Flow helps reduce regional inequality by informing decisions in data-poor regions, learning from data-rich ones. Interestingly, the typologies of urban sprawl can help explain the cross-city model generalization capability. The temporal transferability proves that human dynamics of cities and the process of urbanization can be well captured from the observed built environment by remote sensing.

Cities are complex adaptive systems of spatial networks with vitality generated from continuous human activities and human-environment interactions. Human mobility, as a central component of human activities^{1–3}, reflects the flow of people within and between cities, and plays an important role in transportation management^{4–7}, urban planning^{8–10}, infectious disease modeling^{11–14}, disaster response^{15–18}, and energy production and consumption^{19–21}, thereby promoting policies to make cities resilient and sustainable^{22,23}.

For more than a century, researchers have proposed various models to model human mobility and infer the amount of human movement flows between locations^{24–26}. These models can be classified

into two categories based on their approaches: (1) mechanistic methods and (2) machine learning methods²⁷. Mechanistic methods summarize the human movement process into the interaction mechanism between variables (e.g., population and distance) from the perspective of statistical physics, represented by three popular models at the collective scale: Gravity Model^{1,28}, Intervening Opportunities Model^{29–31}, and Radiation Model³². They mostly rely on probabilistic or parametric fitting methods, which may oversimplify the human movement process and make it difficult to capture complex patterns in the data, thus limiting the accuracy of the model. In recent years, with the rapid increase in urban population, human movements and spatial

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interactions have become more complex. Correspondingly, a couple of machine learning and deep learning models^{33–36} have been developed to unfold the underlying human mobility patterns from historic data and predict future origin-destination (OD) flows in various applications. Existing research shows that graph neural networks (GNNs) have a natural applicability for spatial interaction modeling, such as mobility flows^{37–43}. For the graph construction of static OD flows, the nodes usually represent the geographic units, and the edges can be constructed based on geographical adjacency⁴², transport connectivity^{5,41}, historical OD flows^{36,43}, distance or other types of spatial relations^{44,45}. For the dynamic OD flows, spatial-temporal graph neural networks (STGNNs)^{40,46} have been applied to consider spatial and temporal dependencies to update node features when the topological structure of the network remains unchanged. When the nodes and edges of the network change over time, dynamic graph neural networks^{47–49} capture the network's topological evolution and simulate information interactions. Our study focuses on static OD flow prediction (i.e., travel demand estimation) from urban spatial contexts represented by remote sensing imagery. While prior deep learning models demonstrate superior performance, most of them rely on detailed urban socioeconomic data (e.g., population demographics), land use, points of interest (POI) or mobile phone location data, have not well integrated with other geographical contexts, especially the visual characteristics and spatial arrangement patterns of the built environment (which can be learned from remote sensing data), limiting their applications in data poor regions.

Thanks to the widespread technological advancements, digital connectivity achieved by smart devices and the Internet plays an increasingly vital role in our everyday lives and has fundamentally changed the way people communicate and live^{25,50}. The resulting massive digital datasets, including the trajectories of Global Positioning System (GPS) devices, mobile signal data, and geotagged social media posts, have been widely studied at the collective level, providing a comprehensive understanding of human movements and spatial interactions. This enables more informed policy-making^{12,51}, efficient resource allocation⁵², and an enhanced quality of life in urban areas⁵³. But this has also raised public concerns about privacy issues⁵⁴. Due to privacy protection restrictions, individual-level digital data like human movement trajectories from mobile phones is difficult to obtain directly, especially under the requirements of large-scale and long-time series. Therefore, research on human mobility flow prediction and generation at the collective level holds particular significance in addressing the data gap by inferring human mobility patterns from geographical contexts in cities²⁵.

The physical attributes of terrestrial objects and the heterogeneity of geographical environment containing massive information within urban infrastructure, land cover, and urban form significantly affect and are reciprocally affected by people's social and professional life^{55–57}. Remote sensing-based earth observations provide long-term, continuous and rich geographical landscape information, with the advantages of low acquisition cost and large-scale spatial coverage. Leveraging vast datasets and prior knowledge, there is a great potential to explore the quantitative relationship between geographical environment and human mobility based on remote sensing, which can support the reproduction of statistical patterns of human mobility and reveal the underlying mechanisms of human mobility under various geographical contexts, thus promoting a better understanding of human society through observation of the geographical environment. Existing research has demonstrated the feasibility of using remote sensing imagery for population mapping⁵⁸ and model human activity volumes⁵⁵. However, there still lacks effective methods to predict human mobility flows directly from satellite imagery without any historic mobility data while ensuring accuracy and generalizability.

To this end, in this research, we design a deep learning approach to human mobility flow prediction purely based on satellite imagery,

named as *Imagery2Flow*, so as to model the quantitative relationship between geographic environments (spatial contexts) and human mobility flows by constructing and solving nonlinear problems, and achieve accurate estimations of human mobility flows in urban environments. The proposed *Imagery2Flow* framework uses a self-supervised deep learning approach to encode satellite images into high-dimensional vectors (i.e., embeddings). Then, a graph attention network uses neighbor information aggregation and assigns different weights based on spatial interaction patterns to infer the OD flows. *Imagery2Flow* demonstrates a good performance and flexible spatial and temporal transferability on the top-10 largest Metropolitan Statistical Areas (MSAs) of the United States (U.S.). Through exploring the correlation to urban morphology and heterogeneous effects on model accuracy, we investigate various factors influencing human movements, thus enhancing our comprehension of the complex interactions between human mobility and urban environments and offering insights on urban planning and transportation infrastructure management.

Results

Architecture of Imagery2Flow

As shown in Fig. 1a, the top 10 largest MSAs of the United States are selected as the study areas. Each region of interest (ROI) is partitioned into N smaller geographic units, whose boundaries can be defined by administrative divisions (e.g., census tracts) or square grids. In this study, an ROI is a metropolitan statistical area (MSA), and the census tracts are the smaller geographic units for predicting OD flows. That is, there are N census tracts in each ROI. The flow y_{ij} between area i and area j denotes the number of people moving from area i to area j for commuting, traveling, or other reasons. Generally, the human mobility OD flow prediction task is defined as follows: given the visual characteristics of the urban environment in satellite imagery and the observed spatial structure of flows between any two geographic areas, the model is trained to predict the unknown OD flows within unseen regions.

To predict the human mobility flows within each urban area, *Imagery2Flow* consists of three modules: spatial context embedding, spatial interaction learner, and OD flow predictor (as shown in Fig. 1c). First, it extracts visual features from satellite images as the characteristics of the study areas. For an MSA, the fully covered remote sensing imagery is split into N sub-areas according to the boundaries of each spatial unit. Given a satellite image v_i of geographic area i , the image embedding r_i is obtained to represent the visual features of terrestrial objects within the area in a self-supervised contrastive learning framework. The data augmentations, including random crop, resize, and color distort, are applied to each satellite image v_i twice in an iteration to get two similar satellite views for the same geographic unit, which helps the self-supervised contrastive representation learning for spatial contexts⁵⁹. Then, a deep neural network (e.g., Convolutional Neural Networks, CNN or Vision Transformers) is adopted as the backbone model to encode the satellite views into embedding r_i , then a two-layer nonlinear transformation is employed as a projector to project all the satellite views to a latent space, in which the contrastive loss is calculated to all views for maximizing the embeddings' agreement from the same area. After the contrastive learning process, each geographic area has gained a satellite image neural embedding as a latent representation of its complex visual features.

Second, in order to consider the geographical location, distance, and adjacency influences when predicting the OD flows, we represent the N sub-areas within each ROI into a graph structure to learn their spatial interactions. This graph is constructed based on geographical adjacency, with distance as edge weight. We employ a Graph Attention Network (GAT) as the spatial interaction learner to consider the importance of neighboring nodes with weighting factors, allowing the model to automatically capture geographic areas that have a large or

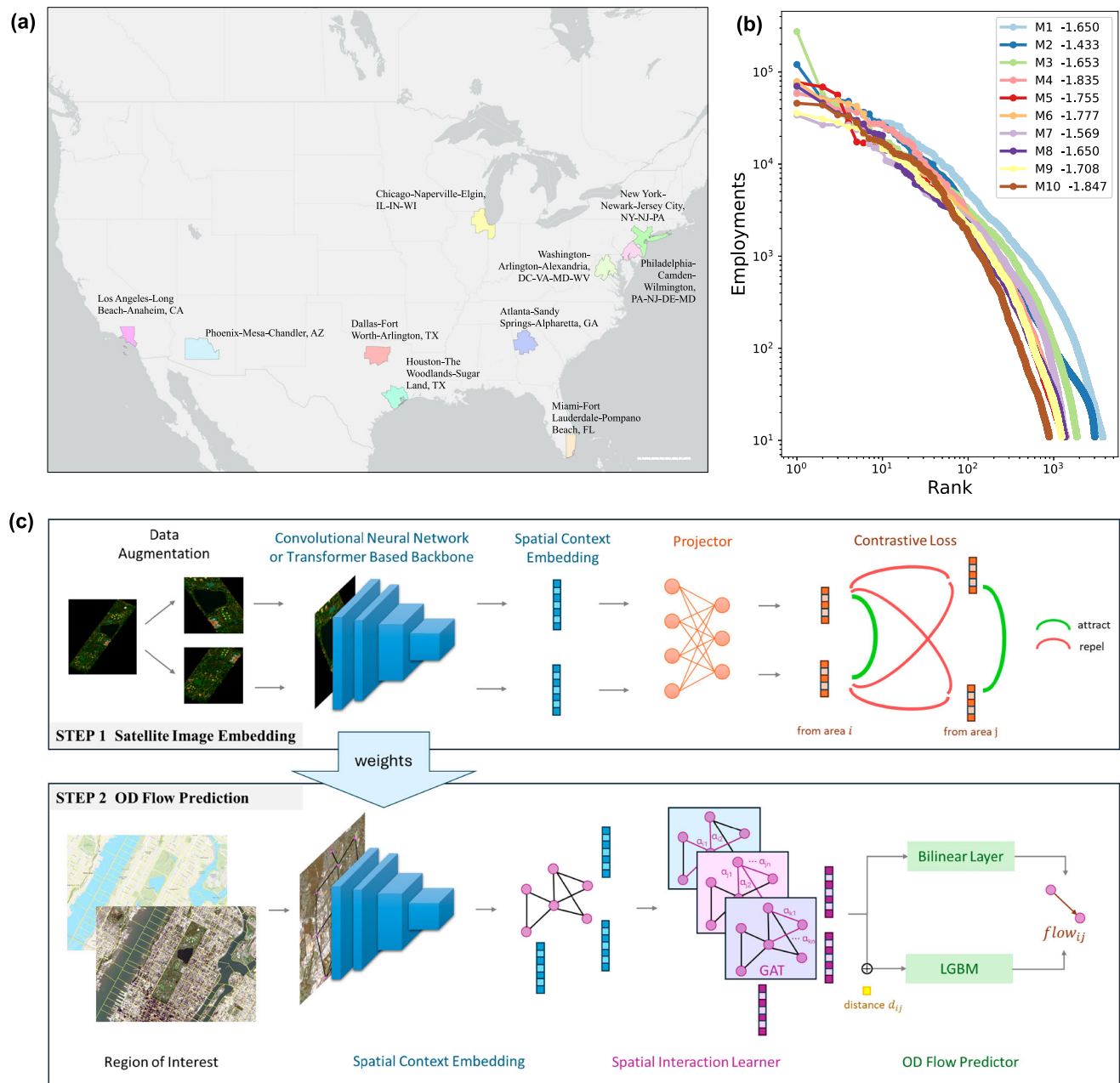


Fig. 1 | Overview of the human mobility flow prediction architecture in Imagery2Flow. **a** The top 10 largest Metropolitan Statistical Area (MSA) of the United States (U.S.), denoted as M1 to M10, corresponding to the following: New York-Newark-Jersey City, NY-NJ-PA (M1); Los Angeles-Long Beach-Anaheim, CA (M2); Chicago-Naperville-Elgin, IL-IN-WI (M3); Dallas-Fort Worth-Arlington, TX (M4); Houston-The Woodlands-Sugar Land, TX (M5); Washington-Arlington-Alexandria,

DC-VA-MD-WV (M6); Philadelphia-Camden-Wilmington, PA-NJ-DE-MD (M7); Miami-Fort Lauderdale-Pompano Beach, FL (M8); Atlanta-Sandy Springs-Alpharetta, GA (M9); and Phoenix-Mesa-Chandler, AZ (M10). **b** The distribution of the number of jobs versus rank for the ten MSAs, which follows the Zipf's law. The legend labels of M1 to M10 are the same as Fig. 1a. **c** Architecture of the Imagery2Flow deep learning model.

small impact on human mobility flows in or out of the current geographic unit. The satellite image embedding r_i is used as initial features of each node (census tract) i , which offers important information about the spatial contexts of each geographic unit. Following the GAT encoding, the node representations are updated by incorporating structural knowledge and neighborhood information through graph representation learning^{39,60}, facilitated by message passing and aggregation mechanisms. The updated node embedding is used as the feature vector of each census tract and fed into the decoder for OD flow prediction in the last module^{40–43}. The satellite image embeddings encode comprehensive urban contexts of each geographic unit, while the spatial interaction learner helps the model understand the spatial

interaction patterns between nodes. Besides, satellite imagery embeddings are transformed and shared among neighboring nodes. In this process, the graph structure is assimilated, which helps to iteratively improve the node embeddings and the relationship with other nodes, while aiding in capturing both local and global information.

Moreover, we utilize the Batch-based Monte-Carlo Loss (BMC Loss)⁶¹ to address the data imbalance issue to achieve better performance than the mean-square-error (MSE) loss. In addition, the mini-batch sampling strategy is utilized in the training process to make the graph represent learning inductive, instead of transductive, by sampling a certain number of nodes as a subgraph in each iteration while training. The nodes in the test set do not appear in the training set as

the origins of the OD flows. This ensures the transferability of the model and allows its application to unseen nodes to infer their corresponding OD flows.

Lastly, a bilinear transformation decoder⁶² as the OD flow predictor is employed in *Imagery2Flow* to learn the spatial interaction patterns from the GAT encoder during training and finally estimate the OD flows between nodes (i.g., census tracts). In addition, we also consider a model variant named *Imagery2Flow-LGBM* by replacing the bilinear decoder with a Light Gradient Boosting Machine (LGBM) to demonstrate the flexibility of our modeling framework, as recent OD flow prediction works using graph neural networks also utilize machine learning decoders^{40,42,43}. We will compare the performance with different baseline models with and without the machine learning decoder.

Experiments and model performance

Using the *Imagery2Flow* framework, we have conducted intensive experiments on human mobility OD flow prediction in the top-10 largest MSAs of the U.S., to provide a comprehensive evaluation of our proposed models. The Census Tracts (CTs) in the MSAs are used as the spatial units for OD flow prediction. The commuting flows among census tracts from the 2020 Origin-Destination Employment Statistics (LODES) dataset in each MSA are utilized as the ground-truth data. The data contains the home and employment locations of workers and their commuting flows. We removed the flows with fewer than 10 commuting workers between census tracts due to their insignificance in accurately capturing stable movement patterns between locations. We use Sentinel-2 and Landsat-8, the two widely used, openly accessible, and stable data sources of satellite imagery with global coverage, with spatial resolutions of 10-meter and 30-meter, respectively, to train deep learning models separately to compare performance. To evaluate the effectiveness of *Imagery2Flow* in predicting mobility OD flows, we employed a wide range of models as baselines, including traditional physical/mechanistic models, machine learning models, and deep learning models. The details of baselines and evaluation metrics are discussed in the 'Methods' section. Each model is independently applied to 10 MSAs. In each MSA, stratified sampling is conducted based on the population in each area, with census tracts divided into a 60% training set, 20% validation set, and 20% test set. The human mobility flows whose origins belong to the training set are involved in the loss computation and the gradient update in the training process but not in the testing process. The 5-fold cross-validation is utilized to get the prediction covering the entire city. For example, in the MSA of New York-Newark-Jersey City, NY-NJ-PA (M1), there are 120,559 flows across 4953 census tracts, with a mean flow intensity of 22.3 and a median of 16. In order to reduce the skewness of the data sampling and improve the performance of the model, we performed a log transformation on the flow intensity during the training of *Imagery2Flow*⁴³.

To illustrate the generalizability of *Imagery2Flow*, we conducted experiments on different backbones for visual feature extraction from satellite imagery in the M1 MSA. The results are shown in the Supplementary Table 5, suggesting that vision transformers perform slightly better than the CNN backbones. Therefore, we utilized the ViT-L/16 backbone for subsequent experiments. Additionally, to assess the effectiveness of each module in the *Imagery2Flow*, we added an ablation experiment. The results are shown in Supplementary Table 4. Specifically, we replaced the visual features extracted from satellite imagery with numerical socioeconomic features (Supplementary Table 3) extracted from OpenStreetMap points of interest (POI) data. The results show that the visual features extracted from Sentinel-2 10-meter satellite imagery have a better performance than the POI data, no matter whether the decoder is a bilinear layer or LGBM. Additionally, compared to the OpenStreetMap data, remote sensing images have a better data quality regarding the completeness and global spatial coverage⁶³ and clear timestamp information, which can ensure

that the model has the potential to infer up-to-date human mobility OD flows globally when statistical or survey data is not available. In addition, there is a small proportion of pairs of OD flows in the dataset accounting for a greater flow intensity; such an observed scaling phenomenon is consistent with other studies using various human mobility datasets in different cities^{3,64,65}. The sparsity of large mobility flow pairs and the presence of data skewness pose significant challenges to the model's performance. The BMC Loss is employed to address this issue by learning the distribution of flow intensity through a cost function analogous to contrastive loss, which outperforms the conventional Mean Square Error (MSE) loss.

Table 1 compares the performance of the models in the top-10 largest U.S. MSAs. Compared with machine learning based and deep neural network-based methods, traditional mechanistic methods, such as the Radiation Model and the Gravity Model, provide worse performance in terms of the Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Common Part of Commuters (CPC) metrics. As a parameter-free model, the Radiation Model is more suitable for long-distance migration modeling for spatial units with coarser granularity³². The error indicators of Gravity Model exhibit fluctuations across different cities, suggesting that the performance is influenced by the metropolitan size and spatial structure. The XGBoost and LGBM achieve impressive performance, demonstrating the applicability of tree-based gradient boosting methods to OD Flow prediction, which is essentially a regression task, and the superiority of visual features extracted from satellite imagery for characterizing spatial contexts in predicting mobility flows. As an improved version of the Gravity Model, the Deep Gravity model³⁵ has non-linearity and multiple hidden neural layers in a fully connected network structure, so using population (Deep Gravity-P) or visual features (Deep Gravity-V) achieves better results than mechanistic models. However, Deep Gravity-V is obviously better than Deep Gravity-P, by incorporating the visual features of complex urban environments from remote sensing imagery. Additionally, the results indicate that Deep Gravity-V-10m using higher resolution sensing imagery (i.e., 10-meter) outperforms Deep Gravity-V-30m (i.e., 30-meter spatial resolution) by capturing richer spatial contextual information. Also as deep learning models, SIGCN³⁶ and GMEL⁴² use graph neural networks for feature encoding, which are distinguished from the fully connected layers of Deep Gravity-V. By replacing the bilinear layer with a gradient boosting regressor after the graph embedding training and fine-tuning process, GMEL narrows the RMSE gap with our *Imagery2Flow-LGBM* to as low as 2.8% when using 10-m resolution imagery, whereas Deep Gravity-V and SIGCN have a gap of 23.5% and 21.8% respectively. Our proposed *Imagery2Flow-LGBM* model demonstrates superior performance on most evaluation metrics, utilizing 10-meter imagery in nine other MSAs and 30-meter imagery in Miami-Fort Lauderdale-Pompano Beach, Florida (M8). The widely used CPC metric represents the degree of correct prediction in OD flow matrices and is more representative of the predicted mobility flow patterns than other metrics³⁵. Compared with traditional machine learning models that directly use visual features, our *Imagery2Flow-LGBM* model uses a graph neural network to further encode the relationships between geographic units, aggregate their neighborhood information, and learn spatial interaction patterns, thus achieving better performance. Specifically, using 10-meter resolution of Sentinel-2 imagery, the RMSE has reduced by a range of 4.5% to 13.9%, MAE by 1.3% to 10.8%, and CPC has increased by 1.3% to 2.6%. The Deep Gravity model is based on a singly constrained gravity model that uses the total outflow volume of each origin area as a known restriction during the OD flow estimation process. Throughout the evaluation of model performance and transferability, such high performance cannot be achieved when the outflow volume is unknown. Similarly, SIGCN builds graphs based on known flows, which restricts its generalizability. When transferring the model to an unseen city or MSA, the *Imagery2Flow* model and its variant *Imagery2Flow-LGBM* can be directly applied to

Table 1 | Performance metrics of different models in the top-10 largest Metropolitan Statistical Areas (MSAs) of the United States

models	M1			M2			M3			M4			M5		
	RMSE	MAE	CPC	RMSE	MAE	CPC	RMSE	MAE	CPC	RMSE	MAE	CPC	RMSE	MAE	CPC
Radiation Model	50.44	25.61	0.16	48.22	27.87	0.11	68.45	31.73	0.16	53.76	28.38	0.13	58.33	29.71	0.13
Gravity Model	50.58	24.72	0.02	94.28	33.58	0.05	123.60	37.82	0.02	135.98	37.74	0.03	135.98	37.74	0.03
XGboost-10m	14.33	7.69	0.83	17.05	7.82	0.84	25.26	10.27	0.81	17.60	9.62	0.81	23.48	11.33	0.78
XGboost-30m	14.46	7.83	0.82	16.88	7.89	0.84	25.70	10.40	0.80	17.16	9.61	0.81	23.43	11.50	0.78
LGBM-10m	14.07	7.69	0.83	15.89	8.22	0.83	23.78	10.19	0.81	16.78	9.56	0.81	22.65	11.47	0.78
LGBM-30m	14.30	7.79	0.82	16.01	8.28	0.83	24.28	10.34	0.80	16.84	9.63	0.81	22.38	11.48	0.78
Deep Gravity-P	26.60	19.61	0.22	32.69	23.17	0.15	43.40	23.84	0.21	32.25	22.88	0.19	38.19	24.33	0.15
Deep Gravity-V-10m	18.48	12.10	0.67	19.69	12.98	0.69	28.43	16.62	0.58	24.15	15.80	0.58	30.15	18.31	0.51
Deep Gravity-V-30m	17.73	11.65	0.69	19.52	12.82	0.70	28.04	16.72	0.58	24.67	16.00	0.58	30.45	18.25	0.51
SIGCN-10m	18.22	9.11	0.78	22.33	11.12	0.76	33.29	11.95	0.76	21.41	11.07	0.77	28.63	12.67	0.73
SIGCN-30m	17.91	9.07	0.78	21.84	11.06	0.76	31.93	11.91	0.76	21.41	11.10	0.77	28.70	13.65	0.73
GMEL-v-10m	13.83	7.38	0.83	16.25	8.30	0.83	23.69	9.64	0.82	16.45	8.97	0.82	21.09	10.19	0.81
GMEL-v-30m	14.01	7.42	0.83	16.57	8.33	0.83	22.71	9.52	0.82	16.58	9.10	0.82	21.19	10.15	0.81
Imagery2Flow-10m	16.47	8.97	0.79	19.46	10.62	0.78	26.20	11.28	0.78	20.76	10.95	0.77	25.91	11.98	0.76
Imagery2Flow-30m	16.66	9.28	0.79	19.69	10.65	0.78	27.22	11.43	0.78	20.93	11.01	0.77	26.36	12.10	0.76
Imagery2Flow-LGBM-10m	13.44	7.36	0.83	14.90	7.72	0.84	21.74	9.42	0.82	15.54	8.76	0.82	20.48	10.23	0.80
Imagery2Flow-LGBM-30m	13.72	7.49	0.83	15.02	7.77	0.84	22.00	9.48	0.82	15.74	8.87	0.82	20.52	10.31	0.80
models	M6	M7	M8	M9	M10										
Radiation Model	57.53	30.26	0.14	59.45	29.26	0.19	51.58	28.71	0.12	48.63	25.79	0.16	49.22	26.11	0.11
Gravity Model	74.44	29.50	0.01	63.49	28.79	0.06	46.21	26.89	0.01	85.91	32.98	0.05	75.40	28.28	0.03
XGboost-10m	17.25	9.88	0.81	16.14	8.90	0.82	16.46	8.95	0.82	14.08	8.00	0.82	14.88	8.22	0.82
XGboost-30m	17.49	10.03	0.81	15.95	8.88	0.82	15.75	8.82	0.83	14.12	8.04	0.82	14.86	8.22	0.82
LGBM-10m	16.81	9.91	0.81	15.48	8.81	0.82	15.40	8.89	0.82	13.54	7.89	0.83	14.42	8.12	0.82
LGBM-30m	17.10	10.03	0.81	15.58	8.80	0.82	15.26	8.77	0.82	13.76	7.95	0.82	14.63	8.25	0.82
Deep Gravity-P	34.85	24.47	0.15	29.81	21.14	0.24	32.91	23.53	0.14	27.79	20.77	0.16	28.35	20.76	0.21
Deep Gravity-V-10m	23.36	16.03	0.61	23.36	15.58	0.57	25.25	17.05	0.57	22.40	15.45	0.51	22.02	15.34	0.54
Deep Gravity-V-30m	24.73	17.01	0.57	21.86	15.11	0.59	23.19	16.11	0.58	21.94	15.46	0.52	20.77	14.25	0.60
SIGCN-10m	22.39	11.71	0.77	19.48	10.13	0.78	20.68	11.20	0.77	17.27	9.08	0.79	17.26	9.17	0.79
SIGCN-30m	22.53	11.78	0.76	19.39	10.09	0.78	20.62	10.89	0.77	17.18	9.02	0.79	17.55	9.48	0.78
GMEL-v-10m	16.36	9.24	0.83	15.07	8.41	0.82	15.19	8.36	0.83	12.94	7.35	0.84	14.10	7.74	0.83
GMEL-v-30m	16.30	9.31	0.82	15.30	8.56	0.82	15.44	8.45	0.83	13.21	7.49	0.84	14.27	7.88	0.83
Imagery2Flow-10m	19.74	10.85	0.79	19.26	10.20	0.78	18.70	10.29	0.79	17.25	9.20	0.79	16.61	8.90	0.80
Imagery2Flow-30m	20.11	11.17	0.78	18.79	10.14	0.78	18.49	10.20	0.79	17.01	9.15	0.79	16.74	8.90	0.80
Imagery2Flow-LGBM-10m	15.45	9.02	0.83	14.30	8.23	0.83	14.71	8.34	0.83	12.51	7.29	0.84	13.50	7.49	0.84
Imagery2Flow-LGBM-30m	15.60	9.13	0.83	14.62	8.28	0.83	14.38	8.20	0.84	12.66	7.29	0.84	13.61	7.60	0.83

The Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) evaluate errors, with smaller values indicating better performance. The Common Part of Commuters (CPC) with values closer to 1, indicating a better quality of performance. Bold indicates the best metrics, while Underscore denotes the second-highest metrics.

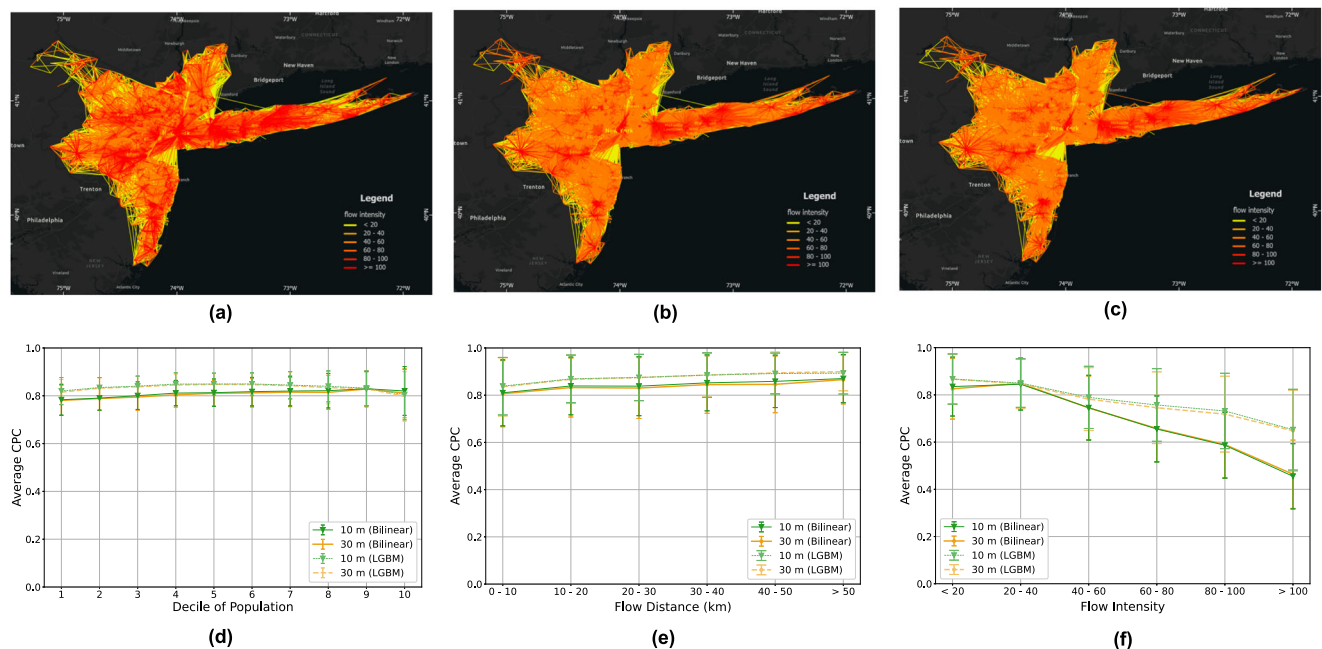


Fig. 2 | Performance of Imagery2Flow and Imagery2Flow-LGBM in New York. There are 120,559 flows among 4953 census tracts. **a.** Real commuting flows. **b.** Predicted flows of Imagery2Flow-LGBM with 10-meter satellite images. **c.** Predicted flows of Imagery2Flow-LGBM with 30-meter satellite images. **d.** Comparison of the performance in terms of Common Part of Commuters (CPC),

varying the decile of the area's population. **e.** The CPC of predicted flows in various distances. **f.** The CPC of predicted flows in various intensities. Error bars represent the standard deviation of CPC of flows across origins, centered at the mean. A 5-fold cross-validation is employed.

the new area with a graph structure between geographic units, requiring only open-source satellite images as input, thus offering greater flexibility.

We further analyze the OD flows predicted by *Imagery2Flow* and *Imagery2Flow-LGBM*, group them according to the population of the origin areas, the intensity and distance of flows, and calculate the average CPC of different groups. As shown in Fig. 2d, we split all geographic units (census tracts) in each MSA into ten equal-sized groups (i.e., deciles), sort them based on their population, as decile 1 accounts for the top 10% of areas by population, and decile 10 accounts for the areas with the smallest population. The model performs well and is stable in every decile. The performance of *Imagery2Flow-LGBM* remains superior in the first nine deciles until slightly declining compared to *Imagery2Flow* in the least populated areas. Recently, points of interest and geo-referenced social media data that can characterize the geographical environment are often used in mobility flow prediction tasks^{35,42,66}. But these data have some limitations of being unevenly distributed across different regions and more abundant in densely populated areas, such as Manhattan, compared to the suburbs of New Jersey in the M1-New York Region, leading to spatial biases in model performance. In contrast, the spectral information capturing terrestrial objects in remote sensing imagery fully covers the entire MSA, exhibiting relatively uniform information richness. Therefore, the difficulty of the model predicting flow does not change much with the population variation across the region.

Based on the distance and flow intensity, we divided the OD flows into six levels, as shown in Fig. 2e and f. Both *Imagery2Flow* and *Imagery2Flow-LGBM* perform well across short and long distances (Fig. 2a–c). However, for high-intensity flows with more travelers, more errors are observed due to the data imbalance, with a large number of low-intensity flows present in the training set. As the intensity increases, the performance of *Imagery2Flow* declines more substantially than that of *Imagery2Flow-LGBM*, which can still achieve the CPC of 0.6 at high flow intensity levels. In addition, there are no significant

differences on the performance of visual features provided by remote sensing images of different spatial resolutions (10-meter or 30-meter).

Correlation to urban morphology

Prior studies of human mobility have investigated the effect of geographical environment on human mobility patterns, such as the distance decay phenomenon, i.e., the reduction in collective human movements or individual displacements as the distance increases between origin and destination. This relationship is commonly modeled using a power law^{24,67} or an exponential form⁶⁸. In addition to terrain morphology and transportation infrastructure, urban form and development also have an impact on human mobility, which can be reflected in satellite imagery. We therefore perform an empirical analysis based on the commuting flows, illustrating the relationship between the predicted mobility and urban morphology (e.g., compactness and centrality).

To quantify the distance decay phenomenon and understand population mobility patterns, we measure the distance distribution of commuting flows. The Akaike weight^{69,70} is utilized to determine which type of distribution is most suitable among exponential, power law, and truncated power law. The results are detailed in Supplementary Note 2 (Supplementary Table 2), which validates that the exponential form approximates the distance decay better than other alternatives in our study. Additionally, the distributions of the predicted flow using the *Imagery2Flow* and *Imagery2Flow-LGBM* models closely match the observed flow, with both the Kullback-Leibler (KL) divergence and Jensen's Shannon (JS) divergence being very small (see Supplementary Table 1). Therefore, the human mobility OD flows predicted by satellite imagery can also capture the information of the urban form and spatial structure that is comparable to the commuting flow. Our proposed method can help urban planners connect satellite imagery to urban morphology from the perspective of human mobility, which has been less explored in prior work, as most existing studies focused on land cover, land use, or population mapping^{55,58,71}.

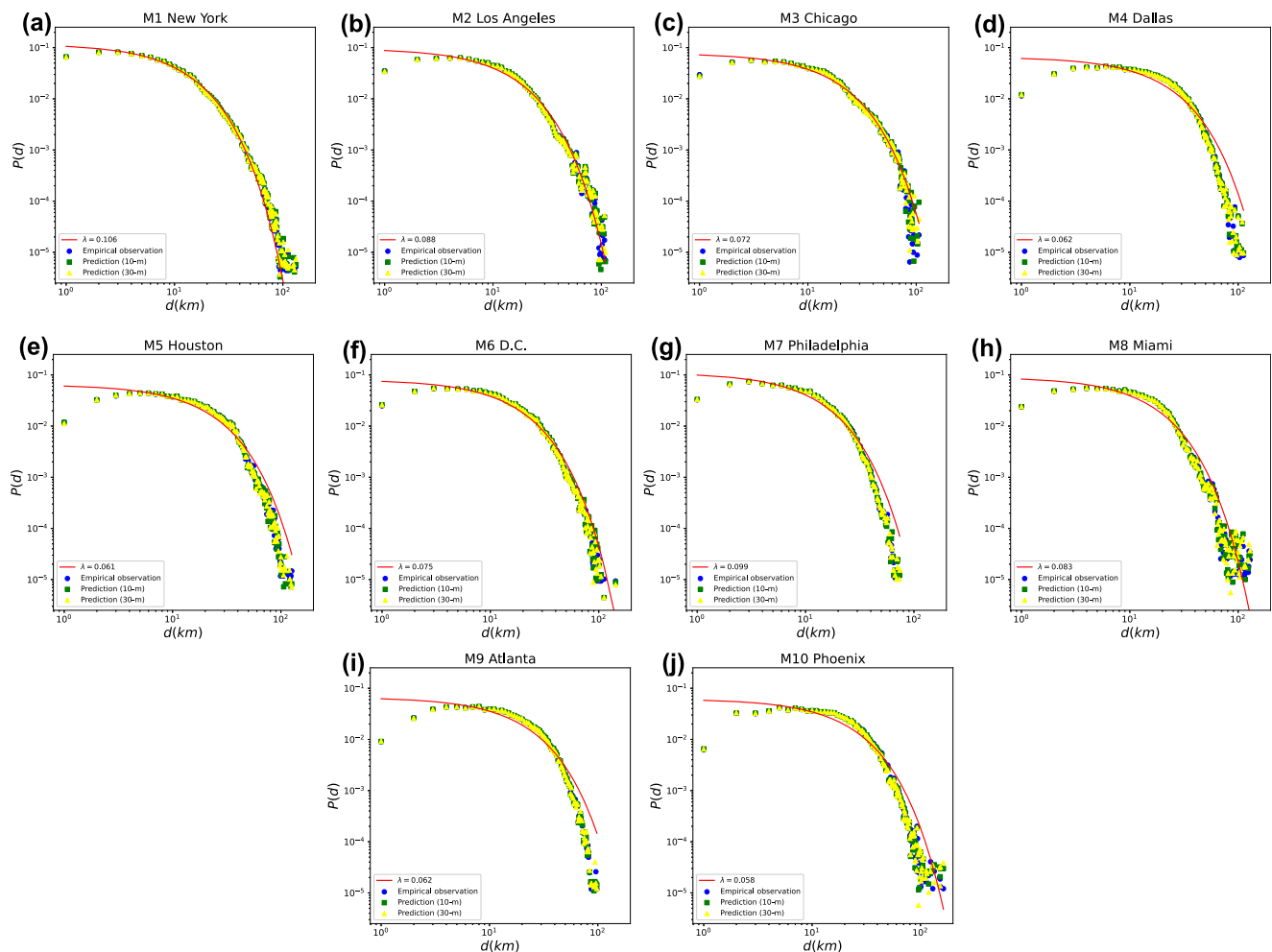


Fig. 3 | The distribution of the observed and predicted human mobility flows by Imagery2Flow-LGBM in the ten United States metropolitan statistical areas.

The red curves represent an exponential fit to the distance distribution of the empirical observations. **a** 120,559 flows in New York-Newark-Jersey City, NY-NJ-PA (M1); **b** 68,712 flows in Los Angeles-Long Beach-Anaheim, CA (M2); **c** 67,090 flows in Chicago-Naperville-Elgin, IL-IN-WI (M3); **d** 57,311 flows in Dallas-Fort Worth-

Arlington, TX (M4); **e** 40,628 flows in Houston-The Woodlands-Sugar Land, TX (M5); **f** 46,261 flows in Washington-Arlington-Alexandria, DC-VA-MD-WV (M6); **g** 45,025 flows in Philadelphia-Camden-Wilmington, PA-NJ-DE-MD (M7); **h** 39,593 flows in Miami-Fort Lauderdale-Pompano Beach, FL (M8); **i** 43,147 flows in Atlanta-Sandy Springs-Alpharetta, GA (M9); **j** 40,443 flows in Phoenix-Mesa-Chandler, AZ (M10).

We fit the mobility flow distance-probability distribution based on Maximum Likelihood Estimation (MLE)⁷⁰ and find that the probability density function of the observed and predicted flows by distance universally exhibit the exponential law $P(d) \sim e^{-\lambda d}$, as shown in Fig. 3. The exponent λ calculated based on the actual OD flow indicates the rate at which commuting distance decays for residents in various cities. A larger λ value signifies that fewer residents undertake long-distance commutes. We further use urban morphology indices that represent compactness and centrality to investigate their correlation with the exponents⁶⁸. The exponent λ shows strong correlation with compactness and centrality of the MSAs, with the Pearson correlation coefficient at 0.6–0.8 (p -value < 0.05) in these ten MSAs (Supplementary Fig. 1). The Boyce-Clark Shape Index (BCSI)⁷² is adopted to quantify the shape irregularity of cities, by projecting n rays from the shape's centroid at equal angular intervals to its boundary and measuring the variation in radial distances. The formula for BCSI is computed by $BCSI = \sum_{i=1}^n \frac{r_i}{\sum_{i=1}^n r_i} \times 100 - \frac{100}{n}$. As the radial distances r_i become more consistent, the BCSI value decreases, indicating a more compact shape. A BCSI value of 0 represents a perfect circle, which has equal radial lengths from the centroid to the boundary, while higher BCSI values represent more irregular shape. For this analysis, we used

36 rays at equal angular intervals from the centroid of each MSA. Supplementary Fig. 1 illustrates a strong positive correlation between the BCSI and λ . In a more compact city, the cost of long-distance commuting is lower due to the more averaged radials, resulting in a higher proportion of long-distance commuters. This is contrary to the correlation obtained in previous research⁶⁸, which showed that in more compact cities, the radius of gyration (ROG) decays more rapidly, that is, fewer residents have a large ROG. There are three possible explanations for this opposing finding. First, the previous research⁶⁸ relies on individual-level mobile phone data, which encompasses a range of individual activities, including daily routines such as home-to-work and entertainment. In compact cities with denser facilities, residents' demand for long-distance movement decreases. However, our research is based on collective commuting survey data, which has fixed activity types and frequencies and is insensitive to urban facilities. This indicates that various types and frequencies of human activities are influenced by urban morphology in distinct ways⁷³. Second, the distance in collective flow data is measured between the centroids of two census tracts, which introduces an aggregation effect on the distance itself. This may differ from the distance decay observed at the individual-level movements. Finally, differences in travel mode preference in different cities and changes

over time may also have influence on individuals' long-distance travel decisions.

Regarding urban centrality factors, this study focuses on population and employment distributions. As population density increases from low to high in different sub-regions, cities shift from a monocentric to a polycentric structure⁷⁴. This transition can be viewed through the lens of urban growth: while higher population densities initially support single central cores, issues such as traffic congestion, resource accessibility challenges, and the demand for diverse economic activities drive the shift towards a polycentric urban spatial structure. For employment, we use the job count of each census tract at the commuting flow destination to create a log-log plot of job distribution versus its rank (Fig. 1b). The slope for each MSA reflects the rate at which the number of employment opportunities declines. A larger negative slope indicates a slower decline in employment, suggesting a polycentric structure with multiple employment centers. A strong correlation is detected between λ and both indices (population density at 0.74 (p -value < 0.05) and employment at 0.70 (p -value < 0.05), Supplementary Fig. 1). The cost of long-distance commuting is lower in a more polycentric city with multiple centers than that in a more monocentric city.

These results prove that the predicted human mobility from remotely sensed data in cities is closely related to urban morphology and spatial structure, by closely resembling the real commuting flow distribution and offering valuable insights. Satellite imagery can help predict the human mobility flows in unseen cities and obtain corresponding distance-decay human mobility patterns under different urban spatial structures.

Spatial Heterogeneity

After analyzing the modeling results, we observed that the prediction accuracy of OD flows varies across different locations, yet these spatial differences are not adequately represented by the metrics in Section 'Experiments and Model Performance'. To gain additional insights from the *Imagery2Flow* model, we explore the spatial heterogeneity of the prediction results across the MSAs. Figure 3 shows the spatial distribution of OD flow prediction errors in New York-Newark-Jersey City (M1), and Supplementary Figs. 2–10 in the Supplementary Note 1 show the results of the remaining nine MSAs. Overall, the *Imagery2Flow* and its variant *Imagery2Flow-LGBM* have higher prediction accuracy around suburban areas in each MSA, while the relatively lower accuracy is found near the urban center. This indicates that the model also has saturation effects from areas with high intensity flows or large populations, where larger errors are more prevalent. This phenomenon is similar to the tasks of estimating population^{75–77}, vegetation indices⁷⁸, carbon emissions⁷⁹ or other variables from remote sensing data sources (including from multi-spectral images and nighttime light data), which is more likely to have larger errors in areas with high values.

Factors that are prone to errors in the OD flow prediction model include differences in the socioeconomics of residents in different neighborhoods^{80,81}, land use and vegetation. We first collect the land use⁸² and land cover⁸³ data to explore predictive accuracy in different groups of census tracts to reveal the patterns between human mobility and urban context. Figure 4c and d present the OD flow prediction accuracy (average CPC) of different groups, which is divided by the dominant land type of the flows' origin areas. As shown in the areas circled in Fig. 4a and b (and Supplementary Figs. 11–19a–b), different factors contributing to the spatial heterogeneity of prediction errors are represented by circles of different colors. The circles labeled 'A' signify areas with vegetation cover, including greenbelts, wildlife management areas, and regional parks. The appearance of vegetation in remote sensing images is very different and the features are complex and diverse, which makes the model perform poorly in areas dominated by vegetation, such as forests and wetlands (indicated in

Fig. 4d). The circles labeled 'B' represent land use types and urban functions, such as recreational areas including encampment, golf and bowling clubs, educational facilities like schools, industrial lands, special types of land such as airfield, commercial zones and landmarks. Buildings on these areas may have similar appearances to residential or workspaces (as indicated in Fig. 4c), leading to overestimation of mobility flows by the model. This is the most common factor in places close to the downtown area in cities. In addition, the circles labeled 'C' signify socioeconomic heterogeneity, such as neighborhood income levels, age of residents, and ethnic demographics. For instance, it is reasonable that the model significantly overestimates the commuter flow in areas with households of median age greater than 55 years old.

Model spatiotemporal transferability

In order to examine the spatiotemporal transferability or generalizability of the model, which is the ability to estimate human mobility flows in other space or time when trained only with data from one specific MSA (we use the largest city name in each MSA as the label of each MSA), we collected the data of top-10 largest MSAs, with different geographies, development histories and lifestyles. Demonstrating the ability of a time- and space-specific model to generalize across diverse regions and timeframes is of considerable importance⁷⁷. Spatiotemporal contexts with limited data availability may benefit from leveraging models trained on contexts with abundant data resources. In addition, it helps to explore and explain different spatial interaction patterns of human dynamics across cities.

The mini-batch sampling strategy facilitates the transferability of our model. To simplify the explanation, each pair "City A-City B" indicates that the model is initially trained on data from city A, and then the learned weights are directly employed on city B's data to predict OD flows in B. In the spatial transfer experiments across all MSAs summarized in Fig. 5, the overall performance of models trained within this MSA (diagonal) is better than that of models trained across MSAs, for training on both 10-meter and 30-meter resolution images. *Imagery2Flow*'s transfer matrix displays notable patterns (Fig. 5a and b), while *Imagery2Flow-LGBM* significantly enhanced the overall transferability (Fig. 5c and d). *Imagery2Flow* employs an end-to-end deep learning framework for inferring OD flows directly from satellite visual features, making its transfer performance inherently dependent on the relevance of these visual features. In contrast, *Imagery2Flow-LGBM* enhances its approach by retraining the LGBM decoder using features that integrate aggregated neighborhood information and spatial distance. We found two pairs "Dallas-Houston" (M4-M5) and "Washington D.C.-Philadelphia" (M6-M7), that performed particularly well while some other metropolitan pairs showed poor spatial transferability results by *Imagery2Flow*. Our experiment results reflect the weak replicability in geospatial AI models under which the model architecture or specification might be generalizable but the model's weights or parameters might be allowed to vary spatially due to the influence of spatial heterogeneity of geographic environments⁸⁴.

Imagery2Flow's transfer performance or generalization capability across regions is affected by land cover type and land use pattern in different areas; the performance correlates with the geographic context similarity. Existing studies have found that green space exposure may help with active transportation (e.g., biking, walking) and the use of public transportation (e.g., subways)^{85,86}, and a lack of vegetation may indicate highways, thoroughfares, etc., which might increase automotive travel and thus link to distinctive human mobility patterns. As shown in Fig. 6, Washington D.C.-Philadelphia (M6-M7) shows the similarity of their geographic contexts, which explains the effective spatial transferability between them to some extent. Most MSAs with extremely low CPC values in flow prediction have different main land cover types, such as Los Angeles-Atlanta (M2-M9), and Miami-Atlanta (M8-M9), etc., on both 10-meter and 30-meter imagery (in Fig. 5a and b).

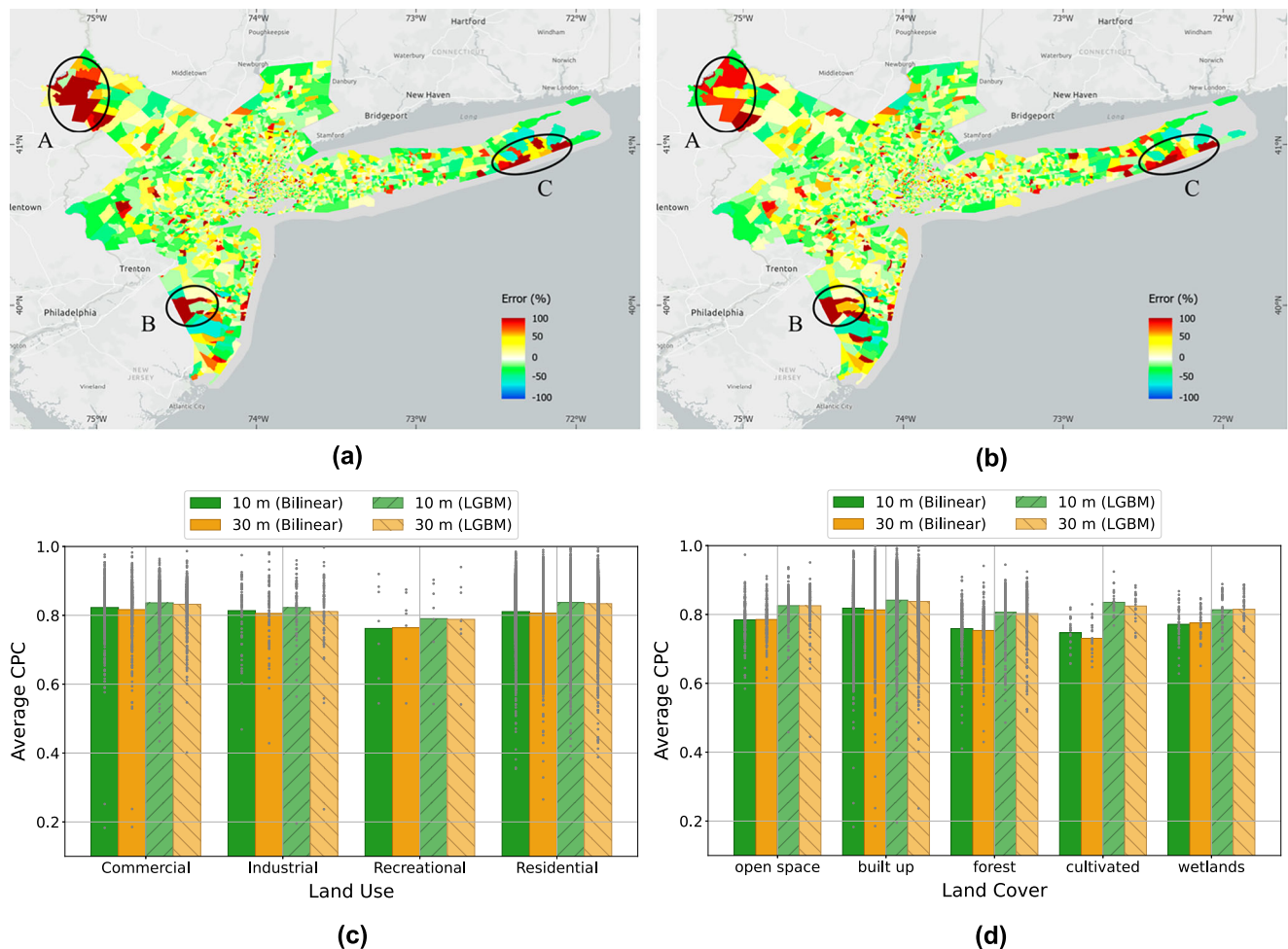


Fig. 4 | Heterogeneous Effect in terms of space and geographical attributes in New York metropolitan area. Spatial distribution of flow prediction errors by *Imagery2Flow-LGBM* according to the origins using **a** Sentinel-2 imagery (10-meter spatial resolution) and **b** Landsat-8 imagery (30-meter spatial resolution), respectively. Note that the labels A, B, and C on the map show the locations with different land covers, land uses, and urban socioeconomic factors. The label A represents areas with vegetation cover, including greenbelt, wildlife management areas, and regional parks; The label B represents areas with diverse land use types and urban

functions, such as recreational areas including encampment, golf and bowling clubs, educational facilities like schools, industrial lands, special types of land such as airfield, commercial zones and landmarks; Label C shows areas with diverse socioeconomic heterogeneity, such as neighborhood income levels, age of residents, and ethnic demographics. **c** Prediction accuracy of different land use groups in terms of Common Part of Commuters (CPC). **d** Prediction accuracy (Average CPC) of different land cover groups. Total 4953 origin census tracts are included in New York.

But the model's spatial transferability still cannot be explained by land cover alone. For "Dallas-Houston" (M4-M5), it works well in places where the main land use types are different. But for "Philadelphia-New York" (M7-M1) and "Washington D.C.-New York" (M6-M1), it doesn't work well with the same main land cover type. This mismatch can be further explained by the spatial forms of population distribution and urban functions in metropolitan areas, which is what urban geographers have long been concerned about and committed to studying. Four distinctive types of residential and nonresidential land use patterns (i.e., typologies of urban sprawl) were identified for all metropolitan areas in the U.S., of which the ten MSAs in this study fell into three⁷¹: (1) *Less-Intensive, Most-Compact, Less-Mixed, Less Monocentric Development*, including Los Angeles (M2), Chicago (M3), Miami (M8), Atlanta (M9), and Phoenix (M10), with little mixing of jobs and housing at a fine-grained scale; (2) *More-Intense, More-Compact, More-Mixed, Polycentric Development*, including New York (M1), Dallas (M4), and Houston (M5), with more fine-grained mixing of jobs and housing; (3) *Least-Intensive, Less-Compact, Most-Mixed, Most-Monocentric Development*, including Washington DC (M6) and Philadelphia (M7), with high degree of fine-grained mixing of housing and jobs. The typologies help explain the model transferability: two metropolitan areas with similar

land use patterns tend to transfer well, as their density of buildings, the spatial arrangement patterns of terrestrial objects, and the jobs and housing matching exhibit high similarity, which is linked to the neighborhood information aggregation and message passing mechanisms in the graph neural networks. Conversely, when urban elements and their spatial patterns diverge, the model transfer effectiveness tends to diminish. The transferability performance is influenced by multiple factors including land cover and land use patterns, population and socioeconomic activity distributions, and spatial structure. The interaction between these factors, combined with data noise, results in variations in model transferability at "A-B" and "B-A", and at different spatial resolutions. The complex quantitative relationship among these spatial contextual characteristics deserves further exploration⁸⁷.

To further verify that the model has truly learned the relationship between urban environments by satellite imagery and human movements, we also applied the models trained on the 2020 imagery to previous years' data spanning approximately 10 years. Since there were no Sentinel-2 images available before 2015, we collected Landsat-5 images from 2010 and 2002 (2004 for M10). As shown in Table 2, the *Imagery2Flow-LGBM* model performance in 2010 and 2002 is good

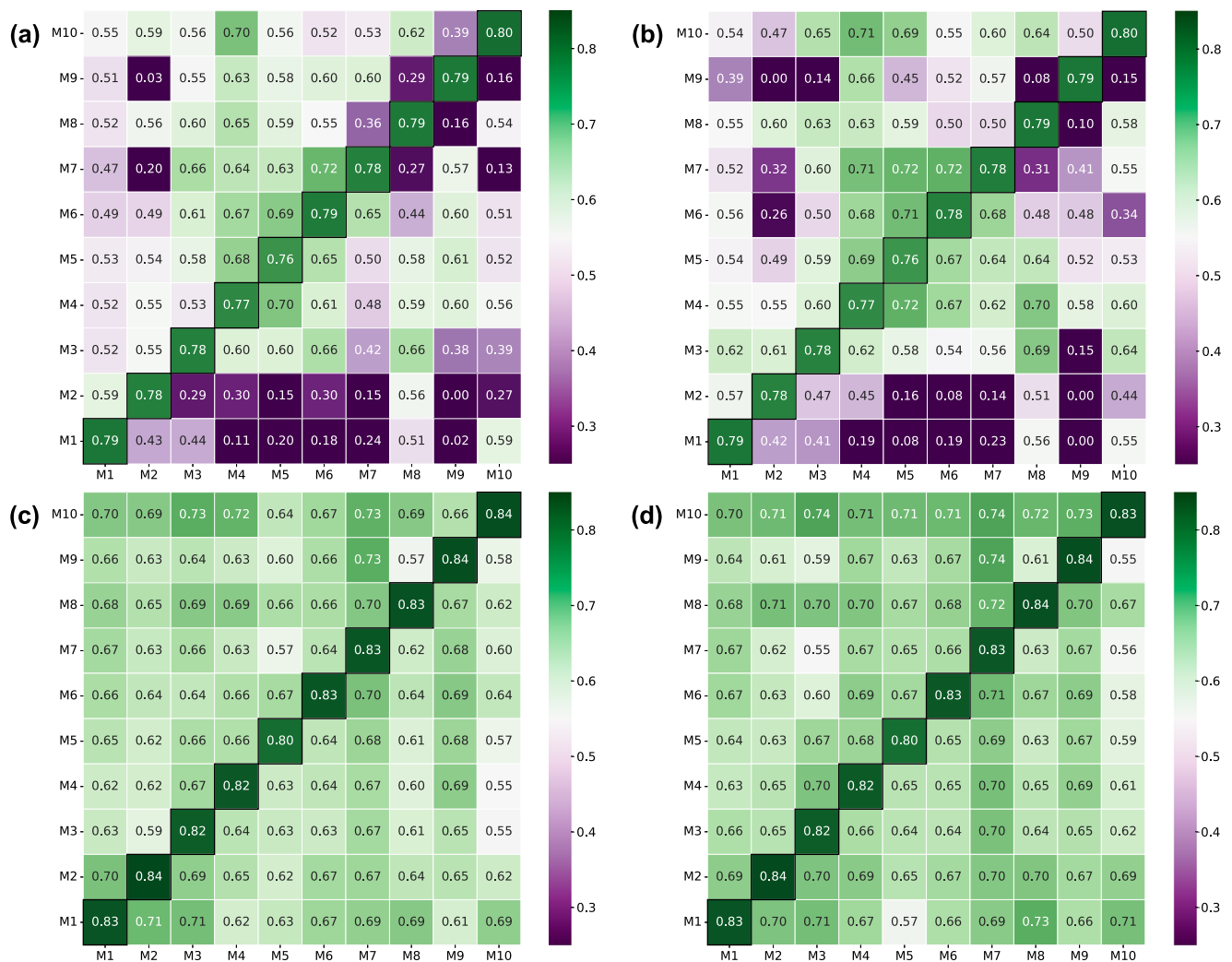


Fig. 5 | Cross-city model performance on mobility flow prediction. The Common Part of Commuters (CPC) values of the *Imagery2Flow* model and the *Imagery2Flow-LGBM* model trained in one Metropolitan Statistical Area (MSA) (on the x axis) and applied to other MSA (on the y axis). **a** Using Sentinel-2 (10-meter) images with bilinear decoder. **b** Using Landsat-8 (30-meter) images with bilinear decoder. **c** Using Sentinel-2 (10-meter) images with LGBM decoder. **d** Using Landsat-8 (30-meter) images with LGBM decoder. M1-M10 represent the ten United States

metropolitan statistical areas: New York-Newark-Jersey City, NY-NJ-PA (M1); Los Angeles-Long Beach-Anaheim, CA (M2); Chicago-Naperville-Elgin, IL-IN-WI (M3); Dallas-Fort Worth-Arlington, TX (M4); Houston-The Woodlands-Sugar Land, TX (M5); Washington-Arlington-Alexandria, DC-VA-MD-WV (M6); Philadelphia-Camden-Wilmington, PA-NJ-DE-MD (M7); Miami-Fort Lauderdale-Pompano Beach, FL (M8); Atlanta-Sandy Springs-Alpharetta, GA (M9); and Phoenix-Mesa-Chandler, AZ (M10).

(with CPC metric above 0.67 across all ten MSAs), which demonstrates very good temporal transferability. Over time, such as years, cities grow as reflected in satellite imagery by the proliferation and densification of buildings and roads, as well as the expansion of urban boundaries, etc. *Imagery2Flow-LGBM* has the temporal transferability to be applied to remote sensing imagery from 10 or even 20 years ago to infer the human mobility flows with good accuracy, proving that this model has learned the underlying relationship between complex geographic contexts in satellite images and human mobility flows.

All the above experiment results indicate that satellite imagery has the ability to reveal human mobility patterns, which can be modeled by the *Imagery2Flow* or *Imagery2Flow-LGBM* models trained at time-stamps and geographic regions with rich data, to predict mobility flows in another time period or other areas with similar spatial contexts where real mobility flows from mobile phone tracking or survey data are unavailable.

Discussion

How to accurately model human mobility in cities is an important research topic in many disciplines such as geographic information

science, transportation, and urban planning^{88,89}. In this research, we propose a deep learning-based framework using satellite imagery to effectively extract geographic contexts and predict the spatial distribution of human movements within cities at high granularity and accuracy. Different from forecasting research on time series, the *Imagery2Flow* and its variant focus on the exogenous mechanism of geographic contexts on mobility OD flow patterns learned from satellite imagery, compared to mechanistic mobility models (e.g., the gravity model and the radiation model) and deep-learning models^{35,36}. The knowledge acquired by the *Imagery2Flow* model not only reveals the impact of spatial heterogeneity of urban characteristics on human mobility but also demonstrates its valuable applicability across different times and locations with similar geographic contexts⁹⁰. Based on the satellite imagery retrieved from public satellite data sources, the *Imagery2Flow* can help municipal leaders and researchers assess the demand for human mobility within a region in a timely and low-cost manner, which is of great significance to multiple practical application scenarios such as transportation infrastructure design, urban planning, epidemic modeling, disaster response, etc.

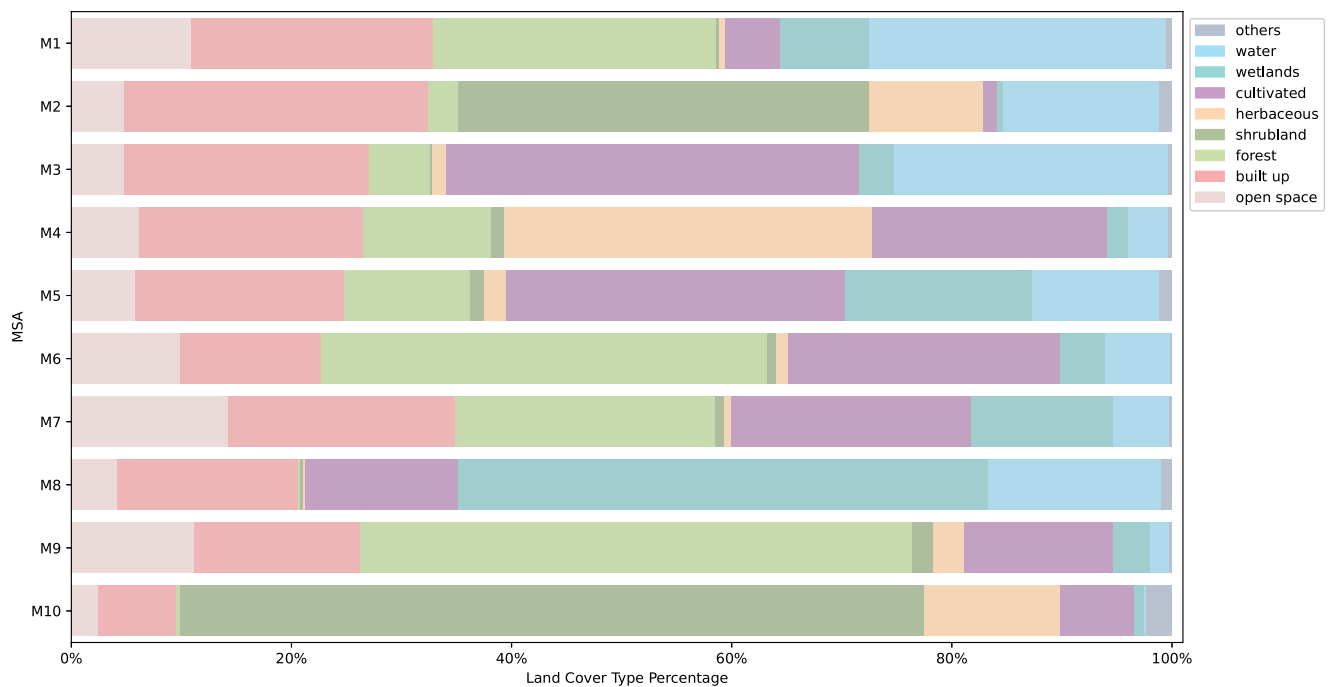


Fig. 6 | Distribution of land cover types in the ten United States metropolitan statistical areas (MSAs). New York-Newark-Jersey City, NY-NJ-PA (M1); Los Angeles-Long Beach-Anaheim, CA (M2); Chicago-Naperville-Elgin, IL-IN-WI (M3); Dallas-Fort Worth-Arlington, TX (M4); Houston-The Woodlands-Sugar Land, TX

(M5); Washington-Arlington-Alexandria, DC-VA-MD-WV (M6); Philadelphia-Camden-Wilmington, PA-NJ-DE-MD (M7); Miami-Fort Lauderdale-Pompano Beach, FL (M8); Atlanta-Sandy Springs-Alpharetta, GA (M9); and Phoenix-Mesa-Chandler, AZ (M10).

Compared to other mechanistic models relying on population or POI data, which are generally more accurate in areas with low population density and fewer trip destinations³⁵, our *Imagery2Flow* performs well in mobility flow prediction in both densely or sparsely populated areas. In highly populated areas, there are many relevant POIs with redundant information, making the mechanistic models harder to predict. Satellite imagery demonstrates a powerful ability in capturing complex and extensive physical properties on the earth, which encompasses natural resources, industrial structures, architectural forms, transportation infrastructure, building density, urban expansion, and so on. These characteristics provide comprehensive contextual information for studying human mobility in cities, while avoiding information redundancy and thus improving performance. Additionally, compared to the POI data, remote sensing images have a better data quality regarding the completeness and global spatial coverage⁶³ and clear timestamp information, which can ensure that the model has the potential to infer up-to-date human mobility OD flows globally when survey data is not available.

Our study offers an alternative perspective on urban morphology analysis by utilizing satellite imagery to examine human mobility patterns. The correlation analysis results indicate that both compactness and centrality are crucial urban factors influencing intra-urban human mobility patterns. By producing high-accuracy predictions of flows from satellite imagery that closely align with actual flows, we can establish connections between earth observations and urban morphology through human dynamics. Moreover, compactness appears to have less influence on human mobility compared to centrality, as evidenced by a smaller Pearson coefficient. Exploring this difference further could provide deeper insights into how various urban characteristics influence human movement and help refine urban planning strategies.

The *Imagery2Flow-LGBM* model trained at one timeframe can directly generate high-accuracy mobility flows using satellite imagery of other timeframes and exhibit good transferability across regions with similar geographic contexts. While there exist diverse historical

backgrounds, demographics, urban planning, and contemporary lifestyles, different cities and metropolitan areas exhibit numerous shared visual features and spatial structures, which can imply human activities and movements. These commonalities suggest underlying patterns of human activities and spatial interactions, transcending geographical and administrative boundaries. Regarding temporal transferability, the models trained based on existing data can be applied to periods where survey data is lacking, helping to restore the spatial patterns of urban population flows in different time periods. At the same time, the spatial transferability of our model ensures that it can be adapted to new, previously unseen urban environments, solving the bottleneck issue in large-scale human mobility data acquisition due to privacy concerns^{26,54}, making it a valuable tool for real-world applications. The weak spatial transferability but strong temporal transferability implies the effects of spatial heterogeneity within human mobility manifested in various contextual characteristics⁸⁴. By leveraging the insights gained from the model transferability and generalizability, we aim to enhance urban sensing, resource allocation, and policy-making processes across diverse urban landscapes, fostering sustainable development and improved quality of life for urban residents globally.

Additionally, our study shows a difference in performance for satellite images with different spatial resolutions, but not as significant as we expected. For in-city models, the 10-meter Sentinel-2 imagery performs better, while the 30-meter Landsat-8 imagery also achieves comparable performance. For across-city models, the performance of region pairs does not necessarily improve with finer spatial resolution. The satellite imagery quality plays a more important role in model performance than spatial resolution. Because each MSA covers a large area, we need to merge multiple images together while ensuring no clouds and shadows. This requirement results in potentially different seasons of imagery capture in the same MSA, causing differences in the presentation of terrestrial objects, thus affecting performance to a certain degree.

In summary, the proposed *Imagery2Flow* and its variant *Imagery2Flow-LGBM* deep learning models achieve promising

Table 2 | Inference performance on other years for the Imagery2Flow-LGBM model trained on 2020 data in each Metropolitan Statistical Area (MSA)

Year	M1			M2			M3			M4			M5		
	RMSE	MAE	CPC	RMSE	MAE	CPC	RMSE	MAE	CPC	RMSE	MAE	CPC	RMSE	MAE	CPC
2010	21.02	10.78	0.75	19.09	10.35	0.77	35.37	14.16	0.70	22.21	12.18	0.73	27.51	13.53	0.72
2002	22.72	11.38	0.73	19.52	10.74	0.75	34.46	14.73	0.69	24.95	13.57	0.69	30.12	14.46	0.67
Year	M6*	M7	M8	M9	M10*										
2010	26.34	14.30	0.68	21.72	11.42	0.74	25.81	13.86	0.67	19.67	10.62	0.75	21.07	10.83	0.75
2002	24.95	13.49	0.69	25.28	12.77	0.71	27.99	14.14	0.67	21.75	11.19	0.73	21.77	11.26	0.73

In M6* (Washington-Arlington-Alexandria, DC-VA-MD-WV), the earliest available data for Washington DC, is from 2010. So the results in 2002 were inferred from remote sensing images and calculated by flows in the other three states within M6. The earliest available data for M10* (Phoenix-Mesa-Chandler, AZ) is from 2004, so the data for 2010 and 2004 were collected and the model applied.

performance in predicting human mobility flows from openly accessible satellite imagery with high accuracy, and potentially is able to generate flows between locations for other metropolitan areas or at various timestamps and geographic regions after training, requiring only the public satellite imagery as input. From a methodological perspective, meta-learning, adaptive learning, knowledge distillation, and other advanced technologies have the potential to facilitate the development of more robust models with enhanced transfer capabilities in the future. In terms of data, the lack of socioeconomic information in satellite imagery may limit its applicability in social sensing⁹¹, thus diminishing the capacity for mobility social interpretation. Satellite images themselves have their own temporal resolution and may reflect monthly, quarterly, annually, or long-term land changes on the earth, such as the proliferation and densification of buildings and roads, as well as the expansion of urban settlements. We acknowledge that short-term variables and self-cyclical effects of human mobility are noteworthy, though they were not the focus of our current work. Different data sources provide varying information, and the insights derived from them need to be tailored to meet specific expectations. For example, social sensors such as smartphones and geo-referenced social media posts contain fine-grained time, location, and human activity information (but are hard to access), making them naturally more well-suited for temporal tracking and dynamic mobility modeling, compared to the satellite imagery. Lastly, due to authoritative OD flow data, we only conducted the experiments in the large metropolitan areas in the United States. Our future work will explore the fusion of remote sensing and social sensing data sources and conduct cross-country experiments at different geographic scales to gain new insights.

Methods

Data sources

All data used in this research are openly available to the public (See ‘Data Availability’ section). The commuting flows data in the U.S is downloaded from the LODES program by the U.S Census Bureau 2020 survey results. The geographic boundary of census tracts is from the U.S Census Bureau Tiger program. The Sentinel-2 Level 2-A, Landsat-8 Collection 2, and Landsat-5 Collection 2 multi-spectral optical pre-processed imagery are used in our study, with a spatial resolution of 10 m x 10 m and 30 m x 30 m, respectively. The satellite imagery is obtained and filtered for clouds and cloud shadows using Google Earth Engine⁹². The travel distances between the centroids of census tracts (driving route in meters) are computed using the Open Source Routing Machine (OSRM)⁹³. For auxiliary data, the land cover for 2020 and land use data for 2015 (the latest available) are sourced from the National Land Cover Database (NLCD) and HISDAC-US.

Satellite image embedding

In order to extract complex visual features of urban environments from satellite images for human mobility flow prediction, we employ an unsupervised machine learning method, which has been proven to have superior performance and strong transferability. We first clip the satellite images of each ROI into N patches according to the division of the spatial areas, and learn a representation vector r_i for each v_i to characterize the visual attributes of terrestrial entities in a self-supervised manner. Due to good performance and wide applicability, we utilize the SimCLR model⁹⁴, a contrastive learning method to learn image embeddings by pulling close the variously augmented views of the satellite imagery from the same geographic area through the optimization of contrastive loss in the latent space.

For the satellite image v_i of area i , a stochastic data augmentation is conducted twice to produce two views of the same area, denoted $\tilde{v}_1$ and $\tilde{v}_2$. The operations of data augmentation are random crop, resize (into a 224×224 dimension), and color distort sequentially. Such data augmentation strategies have been used in existing works for

enhancing spatial scene search⁹⁵. Then we use the ViT-L/16⁹⁶ as the backbone model $f(\cdot)$ to encode the augmented views $\tilde{v}_{i1}, \tilde{v}_{i2}$ into representations r_{i1}, r_{i2} (equation (1)), and a 2-layer nonlinear transformation as the projector $g(\cdot)$ added to the end of the backbone to transform the representations into a latent space, denoted z_{i1}, z_{i2} as a positive pair (equation (2)). The remaining $2(N-1)$ representations of other areas are negative pairs. The normalized temperature-scaled cross entropy loss (NT_Xent)⁹⁷ is used to pull close to the positive pairs and push away the negative pairs in the latent space (equation (3)). We train 120 epochs with the Adam optimizer using a learning rate of 10^{-3} , which decreases by 0.1 at epochs 90 and 110.

$$r_{it} = f(\tilde{v}_{it}) = \text{ResNet}(\tilde{v}_{it}), \quad t \in \{1, 2\} \quad (1)$$

$$z_{it} = g(r_{it}) = W^{(2)} \sigma(W^{(1)} r_{it}), \quad t \in \{1, 2\} \quad (2)$$

$$\begin{aligned} & \text{NT_Xent}(z_{i1}, z_{i2}) \\ &= -\log \frac{\exp(\text{sim}(z_{i1}, z_{i2})/\tau)}{\sum_{k=1}^N \mathbb{1}_{[k \neq i]} \exp(\text{sim}(z_{i1}, z_{k1})/\tau) + \sum_{k=1}^N \exp(\text{sim}(z_{i1}, z_{k2})/\tau)}, \end{aligned} \quad (3)$$

where τ denotes a temperature parameter set as 0.5, $\mathbb{1}_{[k \neq i]}$ is an indicator function whose value is 1 when $k \neq i$. sim is a cosine similarity function.

The above satellite image embedding approach can be applied at different spatial scales. After downloading the satellite imagery for each ROI using Google Earth Engine, the whole imagery for the study area was clipped by the census tract boundaries and resized into 224×224 in the data augmentation step with random crop transformation. When the spatial unit changes to the scale of ZIP Code Tabulation Areas (ZCTA) or even counties, directly resizing the clipped satellite image may result in loss of information. In this case, each ZCTA may be initially divided into smaller square grids, and the remote sensing images are then cropped according to the grids to obtain smaller satellite image patches. The visual features extracted from those image patches within grids can be fused to form the visual features at the ZCTA level. Common methods for feature fusion include summarizing or averaging the visual features of satellite image patches or passing them through multiple multilayer perceptron layers or more advanced feature fusion neural networks⁹⁸.

OD flow prediction using graph attention network

For each metropolitan area (i.e., MSA), a graph is built based on the geographic adjacency of census tracts as nodes. The travel distances between the nodes (using the centroids of census tracts) serve as the edge features of the graph. Using the Graph Attention Network (GAT)⁹⁹ as the spatial interaction learner, we learn the importance between different nodes when aggregating neighbor nodes and edge features during the message passing process, with a bilinear transformation to predict the mobility flow.

Given the state of node $h_i \in R_{m \times 1}$ in the i -th layer and the features of the edge between node i and node j $e_{ij} \in R_{n \times 1}$, the self-attention is conducted as follows:

$$s_{ij} = a(W h_i || V e_{ij} || W h_j), \quad j \in N_i \quad (4)$$

where $W \in R_{k \times m}$ and $V \in R_{t \times n}$ are trainable parameter matrices in the linear transformation to get the message vector passed to neighbors. The concatenated messages are mapped to a real number through $a(\cdot)$ to obtain the attention coefficient s_{ij} between node i and j , which indicates the importance node j 's features to node i . Then attention score α_{ij} is obtained by nonlinear activation σ and softmax

normalization.

$$\alpha_{ij} = \frac{\exp(\sigma(s_{ij}))}{\sum_{k \in N_i} \exp(\sigma(s_{ik}))} \quad (5)$$

where N_i denotes the neighborhood of the i th node in the graph. The message passing involves aggregating information about neighbors and combining it with the feature of the node itself:

$$h'_i = \sigma \left(\sum_{j \in N_i} \alpha_{ij} W h_j + U h_i \right) \quad (6)$$

where $U \in R_{k \times m}$ is a trainable parameter matrix. The resulting h'_i is the new feature of node i output by GAT are aggregated with neighborhood information. To predict the flow from the origin and destination embeddings h_i^{org} and h_i^{dst} from GAT, a bilinear transformation is used as the OD flow predictor (equation (7)).

$$\hat{y}_{ij} = (h_i^{org})^T A h_i^{dst} \quad (7)$$

As discussed above, the OD flow data is unevenly distributed and skewed. Therefore, we performed a log transformation with base 2 when calculating the loss function to predict the value of flow intensity. We utilize the Batch-based Monte-Carlo Loss (BMC Loss)⁶¹ to further address the issue of data imbalance to achieve better performance. BMC Loss leverages all labels in the batch $B = y_1, y_2, \dots, y_K$ as samples, which enables adaptive learning without any prior dependence on the distribution of labels. The equation (8) formulates the loss as a softmax function with temperature $\tau = 2\sigma_{noise}^2$ to control the softness of the probability distribution over targets. In the experiment, we assign the σ_{noise}^2 to the variance of the intensity of the flow in the training set. The form of BMC Loss resembles contrastive loss functions used in self-supervised learning (equation (3)), encourages the model to distinguish between correct and incorrect prediction-target pairs within the batch, thus mitigates the effects of data imbalance.

$$\begin{aligned} \text{loss} &= \text{BMCLoss}(\log y, \log \hat{y}) \\ &= -\log \frac{\exp(-\frac{\|\log y - \log \hat{y}\|_2^2}{\tau})}{\sum_{k=1}^K \exp(-\frac{\|\log y^k - \log \hat{y}\|_2^2}{\tau})} \end{aligned} \quad (8)$$

After the training process described above has been completed, we also train an alternative Light Gradient Boosting Machine (LGBM) as a separate OD flow predictor. The *Imagery2Flow-LGBM* model uses origin and target node embeddings generated by the spatial interaction learner, along with the geographic distance between the nodes, as its input features. The LGBM decoder enhances the overall model performance by capturing complex patterns via tree-ensemble and improves the efficiency by histogram-based learning.

Graph neural networks typically consist of three layers or fewer. Excessive layers can lead to oversmoothing issues¹⁰⁰. After multiple aggregations of neighboring node features, all node features become increasingly similar. We thus trained the *Imagery2Flow* model with 3 layers and embedding size of 512. We trained 100 epochs. In the first 5 epochs, a warmup strategy was applied with an initial learning rate at 10^{-6} , increasing to the learning rate 10^{-5} then decreasing by 0.1 at epochs 40 and 80. The hyperparameters of LGBM decoder include the learning rate, number of estimators, maximum tree depth, sub-sample ratio of the training instance and columns, frequency of sub-sample, L1 and L2 regularization terms on weights. The hyperparameters were searched on the validation set in MI of *Imagery2Flow-LGBM-10m* via Bayesian optimization implemented by Optuna¹⁰¹ and held constant on other MSAs to ensure comparability.

Baselines and evaluation metrics

To demonstrate the superiority of our proposed *Imagery2Flow* and *Imagery2Flow-LGBM* deep learning models, the following well-known human mobility mathematical models are used as baselines and applied to the same data in the ten largest U.S. MSAs.

Radiation Model (RM)³²: The radiation model is a parameter-free model assuming that the flow from location i to j depends on their population m_i and m_j and the population m_{ij} of a circular region around location i whose radius is the distance between location i and j . O_i represents the total number of flows with i as origin.

$$\hat{y}_{ij} = O_i \frac{1}{1 - \frac{m_i}{M}} \frac{m_i m_j}{(m_i + m_j + m_{ij})} \quad (9)$$

Gravity Model (GM)²⁴: The gravity model is the state-of-the-art parametric statistical model for flow prediction or generation task³⁵. By analogy with the law of universal gravitation, it is believed that the OD flows between two places i and j is directly proportional to the attraction and repulsion of the two places (usually represented by population m_i and m_j) and inversely proportional to the distance d_{ij} between the two places, as equation (10a). The deterrence function $f(d_{ij})$ is employed to quantify decreasing effects with distance. The singly constrained gravity model can more accurately estimate mobility flows by fixing the known total outflow quantity, as equation (10b).

$$\hat{y}_{ij} = K m_i m_j f(d_{ij}) \quad (10a)$$

$$\hat{y}_{ij} = O_i \frac{m_i(d_{ij})}{\sum_k m_k(d_{ik})} \quad (10b)$$

Extreme Gradient Boosting (XGBoost)¹⁰² and **Light Gradient Boosting Machine (LGBM)**¹⁰³: XGBoost and LightGBM are highly efficient tree-based gradient boosting algorithms that have gained popularity in regression tasks due to their speed and performance¹⁰⁴. XGBoost highlights its strengths through regularization and level-wise tree growth, enhancing the robustness. LightGBM is known for efficiency and scalability, by binning continuous features and growing trees leaf-wise. For each OD pair, The model input features are formed by concatenating the visual features of Origin and Destination and the distance, which is consistent with the decoder of *Imagery2Flow-LGBM*. The hyperparameters are set to be consistent with those of the decoder of *Imagery2Flow-LGBM*.

Deep Gravity (DG)³⁵: The original input of DG is population and geographical features extracted from OpenStreetMap. To make the results comparable, we define the input of DG into two types, one is population m_i , m_j , and distance d_{ij} , and the other is visual features of remote sensing images r_i and r_j . We denoted the models as Deep Gravity-P and Deep Gravity-V, as equation (11a) and equation (11b) respectively. The suffix represents the spatial resolution of the remote sensing image. The parameters of Deep Gravity-V are determined by grid search.

$$\hat{y}_{ij} = MLP(m_i, m_j, d_{ij}) \quad (11a)$$

$$\hat{y}_{ij} = MLP(r_i, r_j, d_{ij}) \quad (11b)$$

SIGCN³⁶: SIGCN is a graph neural network approach for OD flow prediction. It builds a graph structure based on known flows between nodes. A Graph Convolutional Network (GCN) is used to encode the node features. Parameter settings are determined by grid search.

GMEL⁴²: GMEL is a graph embedding method that learns the representation of nodes on a geo-adjacency graph in a multitask framework. GMEL uses the GAT structure to encode the node features

into the embedding space. Two separate GATs use incoming and outgoing flows as supervisory signals during training, respectively, and integrate their objectives through multi-task learning. After the graph encoding, a Gradient Boosting Regression Tree (GBRT) is trained as a decoder based on the learned embeddings to predict commuting flows. The initial features of the nodes are visual features extracted from remote sensing images. The parameters of the gradient boosting decoder is consistent with those of the LGBM decoder of *Imagery2Flow-LGBM*. The remaining parameters in the graph representation learning and multitask framework are determined by grid search.

The optimal hyperparameters of the deep learning-based baseline models are as follows: the learning rate is set to 1e-4 for DeepGravity-V and GMEL, and 1e-3 for SIGCN; the hidden dimensions are 512 for all three models; the number of GNN layers is 3 for GMEL and 2 for SIGCN.

In addition, the following three evaluation metrics are adopted as OD flow prediction performance measurements. Root Mean Square Error (RMSE) and Mean Absolute Error (MAE): RMSE and MAE are primary performance indicators for regression tasks. They measure the average difference between the predicted values and the ground truth values.

$$RMSE = \sqrt{\frac{\sum_{i,j} (y_{ij} - \hat{y}_{ij})^2}{n}} \quad (12)$$

$$MAE = \frac{\sum_{i,j} |y_{ij} - \hat{y}_{ij}|}{n} \quad (13)$$

Common Part of Commuters (CPC): The prevailing metric utilized for assessing the performance of the mobility flow prediction model is Common Part of Commuters (CPC)²⁴. It's a widely accepted measure for quantifying the similarity between predicted OD flows and the ground truth. The closer the CPC is to 1, the more similar the predicted values are to the true values in OD flow matrices.

$$CPC = \frac{2 \sum_{i,j} \min(y_{ij}, \hat{y}_{ij})}{\sum_{i,j} y_{ij} + \sum_{i,j} \hat{y}_{ij}} \quad (14)$$

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data used in this research are openly accessible to the public. The commuting flows data in the U.S is downloaded from the LODES program by the U.S Census Bureau 2020 survey results, accessible at <https://lehd.ces.census.gov/data/lodes/>. The geographic boundary of census tracts is accessible at <https://www2.census.gov/geo/tiger/TIGER2020/>. The Sentinel-2 Level 2-A, Landsat-8 Collection 2 and Landsat-5 Collection 2 multi-spectral optical pre-processed imagery are used in our study, with a spatial resolution of 10m x 10m and 30m x 30m, respectively. The satellite imagery is obtained and filtered for clouds and cloud-shadows using Google Earth Engine <https://earthengine.google.com/>. The travel distances between the centroids of census tracts (driving route in meters) are computed using the Open Source Routing Machine (OSRM) <https://project-osrm.org/>. For auxiliary data, the land cover for 2020 and land use data for 2015 (the latest available) are sourced from the U.S. Geological Survey National Land Cover Database (NLCD) <https://www.usgs.gov/centers/eros/science/national-land-cover-database> and HISDAC-US <https://dataverse.harvard.edu/dataset/hisdacus>. Other data supporting the figures and the findings of this study are available in the Source data. Source data are provided with this paper.

Code availability

The code of Imagery2Flow is publicly available at GitHub: <https://github.com/GeoDS/Imagery2Flow>. The version of the code used to make the experiments in this paper is available at ref. 105.

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Author contributions

Y.X., S.G., and F. Z. conceived and designed the research. Y.X. collected the datasets, developed the model, and performed the experiments. S.G. and F. Z. directed the study. Y.X., S.G., Q.H., A.G., Q.Z., and F. Z. contributed to interpreting the results, writing the paper and providing feedback on the manuscript drafts.

Competing interests

The authors declare no competing interests.

Additional information

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